GreenWave · Live Demo Script

Team Cubitx · VISTERA 2026 · for the working prototype, about four minutes

0:00 · Home page, before anything is clicked

Good morning. We are Team Cubitx and this is GreenWave: traffic signals that see, decide, coordinate and learn. What you are looking at is not a slide. It is the controller running in this browser. On the right is a live junction: four roads, their queues, and the signal that is deciding right now.

Do: Point at the live junction card. Do not click yet.

0:25 · Click "See the traffic", then Start feed

Everything starts with a camera the city already owns. This is a real crossroads in India, and the AI in the browser is detecting every vehicle live. We split the picture into four regions, one per approach. Green boxes are moving vehicles. Yellow boxes are vehicles that have stopped, and only those count as the queue. So each road's queue on the right is coming straight from the picture.

Do: The feed is already running on the default clip: a CCTV view of a busy Indian crossroads with a signal. Wait for boxes to appear, point at the four region counts, then at a yellow queued box standing at the red.

1:05 · Click "Smarter signals"

The signal reads those queues ten times a second. It holds green while the busy road still has a queue, switches when the other road is longer, and never lets any road wait more than seventy-five seconds. Every decision is explained in plain words, here. Minimum green eight seconds, maximum forty-five, amber two and a half, always.

Do: Read one explanation aloud from the green box, for example "Switched to EW. Reason: the other axis has the longer queue, 9 vs 2".

1:35 · Click "Connected roads"

A junction on its own is not enough. When this junction releases a platoon, it tells the next junction how many vehicles are coming and when. That junction times its green to meet them. That is the green wave, and here it is as a city: J-01 Market Road is this junction, J-02 Central Junction is the next one, four hundred metres down the corridor, and J-03 Hospital Road is the one after. Amber cars are the queues, the signal heads are the live phases, and the cars moving along the corridor are the platoon we just announced.

Do: Point at the platoon moving along the corridor, then click the J-02 card so the footer reads out why its signal changed. Switch to Top view once if you like.

2:00 · Click "Emergency first" (camera trigger, or the button)

Now the moment that matters. An ambulance appears in the camera, and the detector flags it. This junction holds a protected green for it, and in the same second it sends an alert to the next junction with the arrival time. Watch Junction B: the red banner, the countdown, and twelve seconds before the ambulance arrives it clears its own green. Then it acknowledges. The ambulance never meets a red.

Do: Best version: before this step, set Source to Ambulance test clip in See the traffic and Start feed; about ten seconds in the camera flags the ambulance by itself. Fallback: press the ambulance button. Have Junction B visible on the second screen. Read the countdown aloud once.

2:40 · Click "A smoother commute"

Does it actually help? This chart is average waiting time, live. Green is GreenWave, red is the old fixed timer on the same traffic. We can run the comparison on stage: thirty seconds each. Published studies on simulated junctions report forty to forty-seven percent less waiting; this is the same effect on this traffic. Less waiting is less fuel burnt at the stop line.

Do: Press "Run 30 s + 30 s comparison" only if you have the time; otherwise point at the two colours on the chart.

3:10 · Click "Learn every day"

And it learns. At the end of every cycle the junction records how much of the demand each road carried at this time of day and the waiting time it achieved. After two cycles it starts every green from what it learned: the busier road gets a longer minimum green at that hour. This table is the junction's memory, and it is kept between runs, so Monday nine a.m. gets better every Monday.

Do: Point at the highlighted row for the current time slot and the min green column.

3:40 · Scroll to "How the green wave happens", then close

See, decide, coordinate. Those three words are the whole system, and the values under them are live from this junction. In the pilot the camera is the junction's own CCTV, the controller runs on a small box in the signal cabinet, and the messages travel over MQTT on the city network. Nothing you saw today needs new hardware at the roadside. GreenWave, by Team Cubitx. Thank you.

Do: Stop. Take questions.

Likely questions

- Is the camera real? Yes. That is real traffic video and a real neural network detecting vehicles in the browser. A webcam or the junction's CCTV stream plugs into the same slot.
- Is the AI deciding, or rules? The detector and the tracker are neural-network based. The signal decisions today are an adaptive rule set that learns its minimum greens from history. The reinforcement-learning policy, trained in SUMO, is the next phase and replaces the rules behind the same safety guard.
- How does it know a vehicle is an ambulance? A colour signature of the red cross and light bar on the vehicle body, held for four frames, tuned on real ambulance footage. The pilot trains a proper ambulance class and adds siren detection.

- How do junctions talk? Small messages: how many vehicles, when they arrive, and an ambulance flag with its arrival time. Today over a browser channel, in the pilot over MQTT. Same message, same logic.
- What if the camera fails? The guard enforces minimum green and all-red clearance no matter what the detector says. If the feed drops, the junction goes back to its old timer plan.
- Why not just better timers? A timer cannot see traffic. It is wrong the moment demand changes, and it cannot clear a path for an ambulance.

Before you go on

- Open Junction A on the main screen and Junction B on a second screen or a second window before you start, so the alert is visible to the judges the moment you press the button.
- Start the feed about a minute before you present so the learning table already has a row for the current time slot.
- The demo button runs feed, ambulance and the timer comparison by itself if you would rather talk than click.
- Whole script is about four minutes. Split it: one person for See and Smarter signals, one for Connected roads and Emergency, one for Commute and Learn.